All Discrete Operators for the iNS Fractional-Step Solver

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1 Overview and Notation

We solve the incompressible Navier–Stokes equations on an $n_x \times n_y$ staggered Cartesian grid using the Perot (1993) block-LU fractional-step scheme. Six discrete objects must be implemented and verified before the time loop can run:

#	Symbol	Name	Input → Output	Size
1	$\mathbf{G}$	Gradient operator	p-type → u-type	$n_p \to n_u$
2	$\mathbf{D}$	Divergence operator	u-type → p-type	$n_u \to n_p$
3	$\mathbf{bc}_D$	Divergence BC vector	→ p-type	$n_p \times 1$
4	$\mathbf{L}$	Laplacian operator	u-type → u-type	$n_u \to n_u$
5	$\mathbf{bc}_L$	Laplacian BC vector	→ u-type	$n_u \times 1$
6	$\mathbf{A}$	Advection operator	u-type → u-type	$n_u \to n_u$

where $n_p = n_x n_y$ and $n_u = (n_x - 1)n_y + n_x(n_y - 1)$.

Fractional-step pseudo-code (3 lines).

$$\mathbf{R}\,\boldsymbol{u}^F = \mathbf{S}\,\boldsymbol{u}^n + \tfrac{\Delta t}{2}\big(3\mathbf{A}^n - \mathbf{A}^{n-1}\big) + \Delta t\nu\,\mathbf{bc}_L \qquad \text{(Stage 1)}$$

$$\mathbf{D}\mathbf{R}^{-1}\mathbf{G}\,P^{n+1} = \tfrac{1}{\Delta t}\big(\mathbf{D}\boldsymbol{u}^F + \mathbf{bc}_D\big) \qquad \text{(Stage 2)}$$

$$\boldsymbol{u}^{n+1} = \boldsymbol{u}^F - \Delta t\,\mathbf{R}^{-1}\mathbf{G}\,P^{n+1} \qquad \text{(Stage 3)}$$

$$\mathbf{R} \equiv \mathbf{I} - \tfrac{\Delta t\nu}{2}\mathbf{L}, \quad \mathbf{S} \equiv \mathbf{I} + \tfrac{\Delta t\nu}{2}\mathbf{L}, \quad \mathbf{R}^{-1} \approx \mathbf{S} \quad \text{(2nd-order approximation)}$$

Staggered grid index convention. Cell (i, j) is the i-th column and j-th row ($i = 1 \dots n_x$, $j = 1 \dots n_y$).

- $p_{i,j}$: pressure at cell *center*
- $u_{i,j}$: x-velocity at the *east face* of cell $(i-1, j)$ — valid for $i = 2 \dots n_x$
- $v_{i,j}$: y-velocity at the *north face* of cell $(i, j-1)$ — valid for $j = 2 \dots n_y$

The mnemonic: "$u_{i,j}$ is the velocity *entering* cell (i, j) from the left."

2 Gradient Operator G

2.1 Physics and discretization

The gradient maps pressure differences to velocity-face values:

$$(\mathbf{G}\,p)^u_{i,j} = \frac{p_{i,j} - p_{i-1,j}}{\Delta x}, \qquad (\mathbf{G}\,p)^v_{i,j} = \frac{p_{i,j} - p_{i,j-1}}{\Delta y}. \tag{1}$$

This is a *one-sided* (not centered) difference because the u-node sits exactly between two pressure nodes.

Concept

No boundary cases. Every u-node (i=2..nx) has pressure nodes on both sides (i-1 and i). Every v-node (j=2..ny) has pressure nodes above and below. The gradient operator requires **no BC correction vector** — boundary information enters only through **D**, **L**, and **A**.

2.2 Verification

Test G1 (linear field): For $p = ax + by$, the gradient should return a at every u-node and b at every v-node *exactly* (linear functions are reproduced exactly by centered differences).

Test G2 (quadratic field): For $p = x^2 + y^2$, the u-node result should match $2x_{\text{face}}$ to $O(\Delta x^2)$. This probes second-order accuracy.

Test D3 (adjoint): The gradient and divergence are discrete adjoints: $\boldsymbol{u}^\top \mathbf{G}\boldsymbol{p} = \boldsymbol{p}^\top \mathbf{D}\boldsymbol{u}$ (up to machine precision with homogeneous BCs).

3 Divergence Operator D

3.1 Physics and discretization

The divergence maps velocity to cell-center values:

$$(\mathbf{D}\,\boldsymbol{u})_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + \frac{v_{i,j+1} - v_{i,j}}{\Delta y}. \tag{2}$$

This is exact for interior cells. At boundary cells, one or two of the four face velocities are on the domain wall and are *not stored in* $\boldsymbol{u}$.

3.2 Boundary treatment

Concept

The divergence operator is split as

$$\mathbf{D}_{\text{full}}\,\boldsymbol{u} = \mathbf{D}\,\boldsymbol{u} + \mathbf{bc}_D,$$

where **D** acts only on stored interior DOF and $\mathbf{bc}_D$ supplies the missing wall contributions. In the fractional-step solver, $\mathbf{bc}_D$ is pre-computed once and added to the Stage-2 RHS.

Left boundary cell ($i = 1, j = 2 \ldots n_y{-}1$)**:** The face $u_{1,j}$ is a wall velocity ($= u_L^{\text{BC}}$, not in $\boldsymbol{u}$). The Opt_Div function omits this term; BC_Div subtracts $u_L^{\text{BC}}/\Delta x$ from those cells.

Sign convention for $\mathbf{bc}_D$:

Wall	Missing face	Contribution to $\mathbf{bc}_D$
Left	$u_{1,j} = u_L^{\mathrm{BC}}$	$-u_L^{\mathrm{BC}}/\Delta x$
Right	$u_{n_x+1,j} = u_R^{\mathrm{BC}}$	$+u_R^{\mathrm{BC}}/\Delta x$
Bottom	$v_{i,1} = v_B^{\mathrm{BC}}$	$-v_B^{\mathrm{BC}}/\Delta y$
Top	$v_{i,n_y+1} = v_T^{\mathrm{BC}}$	$+v_T^{\mathrm{BC}}/\Delta y$

Corner cells accumulate contributions from two walls simultaneously.

Pressure null space. The operator $\mathbf{DG}$ has a one-dimensional null space (uniform p). We pin $p_{1,1} = 0$ in both the RHS and the operator output during CG iterations.

3.3 Adjoint relationship

A fundamental property of the staggered-grid discretization is

$$\boldsymbol{u}^\top(\mathbf{G}\,\boldsymbol{p}) = \boldsymbol{p}^\top(\mathbf{D}\,\boldsymbol{u}) \quad \text{(up to machine precision with zero BCs).} \tag{3}$$

This is the discrete analogue of the integration-by-parts identity $\int \boldsymbol{u} \cdot \nabla p \, dV = -\int p \, \nabla \cdot \boldsymbol{u} \, dV + \oint p \, \boldsymbol{u} \cdot \hat{n} \, dS$. **Test D3** verifies this.

4 Laplacian Operator L and $\mathbf{bc}_L$

4.1 Physics and discretization

The viscous term $\nu \nabla^2 \boldsymbol{u}$ acts component-wise. For an interior u-node:

$$(\mathbf{L}\,\boldsymbol{u})^u_{i,j} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta x^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{\Delta y^2}. \tag{4}$$

4.2 Homogeneous operator + BC vector decomposition

Concept

As with $\mathbf{D}$, we split the full viscous operator:

$$\mathbf{L}_{\text{full}}\,\boldsymbol{u} = \mathbf{L}\,\boldsymbol{u} + \mathbf{bc}_L,$$

where $\mathbf{L}$ (implemented in `Opt_Laplace_SP`) is *homogeneous* (ignores BC values) and $\mathbf{bc}_L$ (from `BC_Laplace`) carries the Dirichlet wall contributions.

4.3 Ghost-cell boundary treatment

At a wall with Dirichlet condition u_{BC}, we introduce a ghost cell just outside the domain:

$$u_{\text{ghost}} = 2u_{\mathrm{BC}} - u_{\text{interior}}$$

so that the average $(u_{\text{ghost}} + u_{\text{interior}})/2 = u_{\mathrm{BC}}$ exactly. Substituting into Eq. (4) at, say, the bottom edge ($j = 1$):

$$\frac{u_{i,0} - 2u_{i,1} + u_{i,2}}{\Delta y^2} = \frac{(2u_B^{\text{BC}} - u_{i,1}) - 2u_{i,1} + u_{i,2}}{\Delta y^2}$$
$$= \underbrace{\frac{-3u_{i,1} + u_{i,2}}{\Delta y^2}}_{\text{Opt\_Laplace\_SP}} + \underbrace{\frac{2u_B^{\text{BC}}}{\Delta y^2}}_{\text{BC\_Laplace}}$$

The homogeneous operator uses the coefficient -3 (equivalent to using $-u_{i,1}$ in place of the ghost), while $\mathbf{bc}_L$ carries the $2u_{\text{BC}}/h^2$ term.

Table 1: Wall stencil modifications for the u-component. The x-direction stencil is analogous at left/right walls.

Location	Missing node	Homogeneous coefficient	$\mathbf{bc}_L$ contribution
Left ($i = 2$)	$u_{1,j}$	0 (drops term)	$u_L^{\text{BC}}/\Delta x^2$
Right ($i = n_x$)	$u_{n_x+1,j}$	0 (drops term)	$u_R^{\text{BC}}/\Delta x^2$
Bottom ($j = 1$)	$u_{i,0}$	-1 (ghost=-interior)	$2u_B^{\text{BC}}/\Delta y^2$
Top ($j = n_y$)	u_{i,n_y+1}	-1 (ghost=-interior)	$2u_T^{\text{BC}}/\Delta y^2$

Implementation Note

Why -1 at bottom/top but 0 at left/right for u-nodes? A u-node at $i = 2$ sits on a y-face, so the left Dirichlet wall is *half* a cell away and the standard ghost-cell formula is appropriate. The factor-of-2 difference in the BC vector ($u_L^{\text{BC}}/\Delta x^2$ vs. $2u^{\text{BC}}/\Delta y^2$) reflects this asymmetry of the staggered arrangement. Trace through the ghost-cell algebra for each case to confirm.

4.4 Self-adjointness

The homogeneous Laplacian is symmetric: $\boldsymbol{u}^\top \mathbf{L}\boldsymbol{v} = \boldsymbol{v}^\top \mathbf{L}\boldsymbol{u}$. **Test L3** verifies this numerically.

5 BC Vectors: $\mathbf{bc}_D$ and $\mathbf{bc}_L$

These are *constant* (time-invariant for fixed BCs) vectors computed once before the time loop. Their role in each stage of the solver:

Vector	Stage	Role
$\mathbf{bc}_L$	Stage 1 RHS	Adds $\Delta t\nu\, \mathbf{bc}_L$ — wall velocity contribution to viscous term
$\mathbf{bc}_D$	Stage 2 RHS	Adds $\mathbf{bc}_D/\Delta t$ — wall velocity contribution to divergence constraint

Implementation Note

Global mass conservation check for $\mathbf{bc}_D$. Sum the entire $\mathbf{bc}_D$ vector. For a closed cavity with no net inflow the result should be exactly zero. In the lid-driven cavity the top wall moves tangentially (u-direction), contributing nothing to the normal flux, so $\sum \mathbf{bc}_D = 0$. **Test BC1** checks this.

6 Advection Operator A

6.1 Conservative (divergence) form

The advection term in divergence form is:

$$A_x = \nabla \cdot (\boldsymbol{u}\, u) = \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y}, \qquad A_y = \nabla \cdot (\boldsymbol{u}\, v) = \frac{\partial uv}{\partial x} + \frac{\partial v^2}{\partial y}. \tag{5}$$

6.2 Face-interpolation stencils ($\pm\frac{1}{2}$)

Concept

Always use $\pm\frac{1}{2}$ stencils (average to the nearest face). **Never** use ±1 stencils (average across a full cell): this produces direction-dependent truncation errors on a staggered grid and is inconsistent (see lecture slide 49).

For the u-momentum equation at node (i, j):

$$\bar{u}^{\mathrm{x}}_{i\pm\frac{1}{2},j} = \tfrac{1}{2}(u_{i,j} + u_{i\pm1,j}) \qquad \text{(u to left/right x-faces)} \tag{6}$$

$$\bar{u}^{\mathrm{y}}_{i,j\pm\frac{1}{2}} = \tfrac{1}{2}(u_{i,j} + u_{i,j\pm1}) \qquad \text{(u to top/bottom y-faces)} \tag{7}$$

$$\bar{v}^{\mathrm{x}}_{i,j\pm\frac{1}{2}} = \tfrac{1}{2}(v_{i-1,j\pm1} + v_{i,j\pm1}) \qquad \text{(v averaged to top/bottom y-face of u-cell)} \tag{8}$$

The discrete A_x at node (i, j):

$$\boxed{A_x\big|_{i,j} = \frac{(\bar{u}^{\mathrm{x}}_{i+\frac{1}{2},j})^2 - (\bar{u}^{\mathrm{x}}_{i-\frac{1}{2},j})^2}{\Delta x} + \frac{\bar{u}^{\mathrm{y}}_{i,j+\frac{1}{2}} \cdot \bar{v}^{\mathrm{x}}_{i,j+\frac{1}{2}} - \bar{u}^{\mathrm{y}}_{i,j-\frac{1}{2}} \cdot \bar{v}^{\mathrm{x}}_{i,j-\frac{1}{2}}}{\Delta y}} \tag{9}$$

6.3 v-momentum stencil

For the v-momentum equation at v-node (i, j):

$$\bar{u}^{\mathrm{y}}_{i\pm\frac{1}{2},j} = \tfrac{1}{2}(u_{i\pm1,j} + u_{i\pm1,j-1}) \qquad \text{(u straddling two rows, averaged to right/left x-face of v-cell)} \tag{10}$$

$$\bar{v}^{\mathrm{x}}_{i\pm\frac{1}{2},j} = \tfrac{1}{2}(v_{i,j} + v_{i\pm1,j}) \qquad \text{(v to right/left x-face)} \tag{11}$$

$$\bar{v}^{\mathrm{y}}_{i,j\pm\frac{1}{2}} = \tfrac{1}{2}(v_{i,j} + v_{i,j\pm1}) \qquad \text{(v to top/bottom y-face)} \tag{12}$$

$$\boxed{A_y\big|_{i,j} = \frac{\bar{u}^{\mathrm{y}}_{i+\frac{1}{2},j} \cdot \bar{v}^{\mathrm{x}}_{i+\frac{1}{2},j} - \bar{u}^{\mathrm{y}}_{i-\frac{1}{2},j} \cdot \bar{v}^{\mathrm{x}}_{i-\frac{1}{2},j}}{\Delta x} + \frac{(\bar{v}^{\mathrm{y}}_{i,j+\frac{1}{2}})^2 - (\bar{v}^{\mathrm{y}}_{i,j-\frac{1}{2}})^2}{\Delta y}} \tag{13}$$

6.4 BC embedding

At a Dirichlet wall, the missing stencil neighbor is replaced directly by the BC value:

$$\bar{u}^{\mathrm{x}}_{\text{wall face}} = \tfrac{1}{2}(u_{\mathrm{BC}} + u_{\text{interior}}).$$

No ghost nodes are needed. This is in contrast to the Laplacian treatment, where BCs enter through a separate correction vector.

6.5 Properties

Property	Statement
Conservation	Divergence form conserves momentum discretely (telescoping sums).
Energy	$\boldsymbol{u}^\top \mathbf{A}(\boldsymbol{u}) \approx 0$ for divergence-free $\boldsymbol{u}$ (Test A2).
Uniform flow	$\mathbf{A}(U_0\hat{x}) = 0$ exactly (Test A1).
Non-dissipative	No implicit numerical diffusion — oscillations at high Re possible.

7 Conjugate Gradient Solver

7.1 Algorithm

Both Stage 1 and Stage 2 require solving $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ where $\mathbf{A}$ is SPD. The matrix-free CG algorithm is:

Input: function handle `Afun`, right-hand side $\boldsymbol{b}$, tolerance `tol`

1. $\boldsymbol{x}_0 = \mathbf{0}, \quad \boldsymbol{r}_0 = \boldsymbol{b} - \mathbf{A}(\boldsymbol{x}_0), \quad \boldsymbol{p}_0 = \boldsymbol{r}_0$

2. For $k = 0, 1, 2, \ldots$:

$$\alpha_k = \frac{\boldsymbol{r}_k^\top \boldsymbol{r}_k}{\boldsymbol{p}_k^\top \mathbf{A}(\boldsymbol{p}_k)} \qquad \boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$$

$$\beta_k = \frac{\boldsymbol{r}_{k+1}^\top \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^\top \boldsymbol{r}_k} \qquad \boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \mathbf{A}(\boldsymbol{p}_k)$$

$$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$$

3. Stop when $\|\boldsymbol{r}_k\|_2 / \|\boldsymbol{b}\|_2 <$ `tol`.

7.2 Operators passed to CG

Stage	Operator function	Notes
1	`Opt_R`: $\boldsymbol{x} \mapsto \boldsymbol{x} - \frac{\Delta t \nu}{2}\mathbf{L}\boldsymbol{x}$	SPD (positive diagonal dominant)
2	`Opt_T`: $\boldsymbol{x} \mapsto \mathbf{DS}(\mathbf{G}\boldsymbol{x})$	SPD up to null-space pin

Pressure null space. $\mathbf{DG}$ is only *semi*-definite: constant pressure fields are in the null space. We fix $p_{1,1} = 0$ by zeroing entry $(1,1)$ of both the operator output and the RHS before every CG iteration (function `pin_pressure` in the script).

8 Complete Operator Chain: Putting It Together

Listing 1: Full fractional-step loop (schematic)

```
% One-time setup
Opt_R  = @(x) x - (dt*visc/2) * Opt_Laplace_SP(x);
Opt_S  = @(x) x + (dt*visc/2) * Opt_Laplace_SP(x);   % R^{-1} approx
Opt_T  = @(x) Opt_Div(Opt_S(Opt_Grad(x)));            % D R^{-1} G

BC_D_vec = BC_Div();      % constant for fixed BCs
BC_L_vec = BC_Laplace();  % constant for fixed BCs

P = zeros(np,1);   U = zeros(nu,1);   adv1 = Adv(U);

while time < endTime
    adv0 = Adv(U);

    % Stage 1: solve  R uF = S u + dt*(1.5 A^n - 0.5 A^{n-1}) + dt*nu*bc_L
    RHS1 = Opt_S(U) + dt*(1.5*adv0 - 0.5*adv1) + dt*visc*BC_L_vec;
    [UF, it1] = CGSolve(Opt_R, RHS1, 1e-8);

    % Stage 2: solve  T P = (D uF + bc_D) / dt   [with pressure pinning]
    RHS2 = (Opt_Div(UF) + BC_D_vec) / dt;
    RHS2(ip(1,1)) = 0;
    [P, it2] = CGSolve(@(x) pin_pressure(Opt_T(x)), RHS2, 1e-8);

    % Stage 3: project onto divergence-free space
    U = UF - dt * Opt_S(Opt_Grad(P));   % Opt_S = R^{-1}

    adv1 = adv0;
end
```

9 Verification Test Summary

Test	Operator	What is checked	Expected result
G1	$\mathbf{G}$	Gradient of linear p	Exact (machine ε)
G2	$\mathbf{G}$	Gradient of quadratic p	Error $O(\Delta x^2)$
D1	$\mathbf{DG}$	Composition $\approx \nabla^2$	Error $O(\Delta x^2)$ interior
D2	$\mathbf{D}$	Divergence of div-free field	Error $O(\Delta x^2)$ interior
D3	$\mathbf{D}, \mathbf{G}$	Adjoint: $\boldsymbol{u}^\top \mathbf{G}\boldsymbol{p} = \boldsymbol{p}^\top \mathbf{D}\boldsymbol{u}$	Machine ε
BC1	$\mathbf{bc}_D$	Global mass balance $\sum \mathbf{bc}_D = 0$	Exact
BC2	$\mathbf{bc}_L$	Top-wall BC value $= 2U^*/\Delta y^2$	Exact
L1	$\mathbf{L}$	Laplacian of quadratic field $= 2$	Exact interior
L2	$\mathbf{L}$	Laplacian of linear field $= 0$	Machine ε
L3	$\mathbf{L}$	Self-adjoint: $\boldsymbol{u}^\top \mathbf{L}\boldsymbol{v} = \boldsymbol{v}^\top \mathbf{L}\boldsymbol{u}$	Machine ε
A1	$\mathbf{A}$	Uniform flow gives zero advection	Machine ε
A2	$\mathbf{A}$	Energy: $\boldsymbol{u}^\top \mathbf{A}(\boldsymbol{u}) \approx 0$	Small ($O(\Delta x^2)$)
CG1	CG	Solves Poisson problem to tolerance	$<$ tol
FULL	$\mathbf{T}$	$\boldsymbol{p}^\top \mathbf{T}\boldsymbol{p} > 0$ (SPD)	Positive

Task

Exercise 1. Run all tests on a 16×16 grid. Note which tests pass to machine precision and which pass only to $O(\Delta x^2)$. Explain why the distinction arises.

Exercise 2. Repeat on a 32×32 grid. For tests that pass to $O(\Delta x^2)$, verify that the error reduces by a factor of 4.

Exercise 3. Examine `TEST D3` (adjoint check). Replace `Opt_Div` with a deliberately wrong version (e.g. swap + and − signs) and re-run. Does the adjoint test catch the error?

Exercise 4 (challenge). The pressure Poisson operator $\mathbf{T} = \mathbf{D}\mathbf{R}^{-1}\mathbf{G}$ is approximated using $\mathbf{R}^{-1} \approx \mathbf{S}$. Show analytically that this approximation introduces an $O(\Delta t^2)$ error and does not affect the overall second-order temporal accuracy of the scheme.